DIVYANSHU YADAV

Computer Science AI & ML

+91-9671498171 divyanshuyadav715@gmail.com www.linkedin.com/in/divyanshu-yadav-2911a52b2

Education

Vellore Institute of Technology, Bhopal **Sept 2023 – June 2027**
Bachelor of Technology in Computer Science *CGPA: 8.59/10*

RPS Public School, Rewari **April 2021 – May 2022**
Higher Secondary School (Class XII) *91.6%*

RPS Senior Secondary School, Mahendargarh **April 2019 – May 2020**
Secondary School (Class X) *86.6%*

Professional Summary

Versatile software developer with a strong foundation in data structures and algorithms, proficient in Python, C++, C, and Java. Experienced in building diverse applications including machine learning systems and predictive analytics pipelines that drive strategic decisions and business ROI. Equipped with a solid understanding of AWS architecture.

Technical Skills

Languages: C++, Python, Java, HTML, CSS
Developer Tools: Git, GitHub, VS Code
Core Skills: Product Analytics, A/B Testing, Root Cause Analysis, Data Analysis, Machine Learning (XGBoost, SVM), Object-Oriented Programming, Data Structures & Algorithms

Certifications

- NPTEL Cloud Computing
- NPTEL Internet of Things
- AWS Certificate of completion of AWS Training S Certification course
- Coursera – Applied Machine Learning in Python (University of Michigan)

Projects

SynthMarket – Autonomous RL Trading System with GANs | *PyTorch, FastAPI, Streamlit* **2024**
– Built Adaptive Trading Protocol combining Time-Series GANs (for synthetic data) and Proximal Policy Optimization (PPO) for algorithmic trading.
– Implemented MLOps features using real-time KL-Divergence drift detection to trigger kill switches during market anomalies.
– Deployed a FastAPI microservice and Streamlit dashboard with <5ms inference latency and 17.72% simulated average cumulative return.

InferX – Scalable AI Serving Engine | *Java, Spring Boot, Python, Redis, PyTorch* **2024**
– Engineered a distributed deep learning inference gateway separating client API from GPU workers using an event-driven Redis architecture (Pub/Sub).
– Implemented Dynamic Batching at the worker level, grouping incoming HTTP requests into a single tensor block for highly efficient parallel GPU execution.
– Load-tested with Locust to handle 1,000 concurrent users achieving 309 RPS with a p95 latency of 185ms and 0.00% failure rate.

Predictive Customer Churn & Intervention Engine | *Python, XGBoost, SQLite, Streamlit*
– Engineered an end-to-end ML pipeline analyzing 50k+ user behavioral logs to identify at-risk cohorts.
– Bypassed standard purely-predictive models by implementing a custom Customer Acquisition Cost (CAC) vs. Lifetime Value (LTV) routing logic to target retention discounts.
– Projected to reduce wasted retention spend by 30% and save 45 Lakhs in annualized revenue.

E-Commerce Conversion Funnel & Drop-Off Analysis | *Advanced SQL, Python, Tableau / BI Dashboards*
– Simulated and analyzed 100,000+ web traffic logs using SQL window functions and CTEs to identify bottlenecks.
– Built an interactive dashboard to visualize stage-to-stage conversion rates across the entire user journey.
– Identified a critical mobile UI drop-off, recommending a UX redesign projected to recapture 25 Lakhs in monthly revenue.

MediGuide Predictive Healthcare | *C++, Python, Flask, TensorFlow*
– Developed an AI-powered healthcare web application for symptom-based disease prediction and care recommendation using Machine Learning and Flask.
– Engineered the data processing pipeline by cleaning, transforming, and integrating 8 medical datasets into optimized lookup structures for real-time predictions.
– Designed a dynamic doctor recommendation system that maps 41 diseases to appropriate specialists and generates personalized referral links across multiple healthcare platforms.

Co-Curricular Activities

- Managed and coordinated 50+ events across AI club and by collaborating with cross-functional student teams.
- Led planning, communication, logistics, and stakeholder coordination to ensure successful event execution.
- Solved 100+ LeetCode problems, strengthening analytical thinking and structured problem-solving abilities.
- Worked with faculty members, student volunteers, and external participants during event management activities.